Lab 5: Static Code Analysis

Before-and-after analysis of the conference room scheduler using Pylint, Bandit, and Flake8.

Submission date: October 14, 2026

Repository: https://github.com/satvik-2006/L5_12_room_scheduler

Results at a glance

Tool	Before changes	After changes
Pylint	5.60 / 10	10.00 / 10
Bandit	2 findings: 1 high, 1 low	No findings
Flake8	13 violations	No violations

Analysis environment: Python 3.12.14, Pylint 4.1.3, Bandit 1.9.4, and Flake8 7.4.1. The before reports were captured from the untouched starter file. The after reports were captured from the final scheduler.

Generated screenshot evidence

The screenshots below are generated from the saved source files and captured analyzer output. They provide a visual before/after record alongside the full text reports in the appendix.

Source code screenshots

Before

Before - scheduler_system.py (generated source view)

```python
import json
import os
from datetime import datetime

# Global state
reservations = {}
equipment_pool = {"projector": 2, "microphone": 4, "webcam": 1}
conflicts = []
next_reservation = 1

def schedule_room(room_id, time_slot, equipment_list=[]):
    global next_reservation
    for r in reservations:
        if reservations[r]["room"] == room_id and reservations[r]["slot"] == time_slot:
            conflicts.append("%s double-booked at %s" % (room_id, time_slot))
    for e in equipment_list:
        equipment_pool[e] = equipment_pool[e] - 1
    reservations[next_reservation] = {"room": room_id, "slot": time_slot, "equipment": equipment_list}
    next_reservation += 1
    return next_reservation - 1

def cancel_reservation(reservation_id):
    try:
        for e in reservations[reservation_id]["equipment"]:
            equipment_pool[e] += 1
        del reservations[reservation_id]
    except:
        pass

```

After

After - scheduler_system.py (generated source view)

```python
"""Schedule meeting rooms and equipment while recording booking conflicts."""

import json
from collections import Counter
from datetime import datetime
from pathlib import Path


DEFAULT_EQUIPMENT = {"projector": 2, "microphone": 4, "webcam": 1}


def slot_hours(time_slot: str) -> int:
    """Return the duration of a slot such as ``09-11`` in hours."""
    parts = time_slot.split("-")
    if len(parts) != 2:
        raise ValueError("A time slot must use the HH-HH format.")

    try:
        start, end = (int(part) for part in parts)
    except ValueError as error:
        raise ValueError("Time slot hours must be whole numbers.") from error

    if not 0 <= start < 24 or not 0 < end <= 24 or end <= start:
        raise ValueError("A time slot must end after it starts in one day.")
    return end - start


class Scheduler:
    """Manage room reservations, equipment stock, and conflict logs."""

```

Program run screenshots

Before

Before program run

```
PS> python scheduler_system.py

Slot length for b: -2
'cat' is not recognized as an internal or external command,
operable program or batch file.
Slot for Board-2, like 10-12 (Enter for 10-12): Equipment left: {'projector': 1, 'microphone': 4, 'webcam': 0}
Reservations: {1: {'room': 'Board-1', 'slot': '09-11', 'equipment': []}, 2: {'room': 'Board-1', 'slot':
    '09-11', 'equipment': []}, 3: {'room': 'Huddle-3', 'slot': '14-12', 'equipment': []}, 4: {'room':
    'Huddle-3', 'slot': 1530, 'equipment': []}, 5: {'room': 'Board-2', 'slot': '10-12', 'equipment':
    ['projector', 'webcam']}}
```

After

After program run

```
PS> python scheduler_system.py

Invalid booking rejected: A time slot must end after it starts in one day.
Slot for Board-2, like 10-12 (Enter for 10-12): 2026-10-14 12:12:16.075200 Board-1 double-booked at 09-11
Equipment left: {'projector': 1, 'microphone': 4, 'webcam': 0}
Reservations: {1: {'room': 'Board-1', 'slot': '09-11', 'equipment': []}, 2: {'room': 'Board-1', 'slot':
    '09-11', 'equipment': []}, 3: {'room': 'Board-2', 'slot': '10-12', 'equipment': ['projector', 'webcam']}}
```

Pylint screenshots

Before

Before Pylint report

```
PS> pylint scheduler_system.py

************ Module scheduler_system
scheduler_system.py:18:0: C0301: Line too long (102/100) (line-too-long)
scheduler_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
scheduler_system.py:9:0: C0103: Constant name "next_reservation" doesn't conform to UPPER_CASE naming style
    (invalid-name)
scheduler_system.py:11:0: C0116: Missing function or method docstring (missing-function-docstring)
scheduler_system.py:11:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
scheduler_system.py:12:4: W0603: Using the global statement (global-statement)
scheduler_system.py:13:4: C0206: Consider iterating with .items() (consider-using-dict-items)
scheduler_system.py:15:29: C0209: Formatting a regular string which could be an f-string (consider-using-f-
    string)
scheduler_system.py:22:0: C0116: Missing function or method docstring (missing-function-docstring)
scheduler_system.py:27:4: W0702: No exception type(s) specified (bare-except)
scheduler_system.py:30:0: C0116: Missing function or method docstring (missing-function-docstring)
scheduler_system.py:34:0: C0116: Missing function or method docstring (missing-function-docstring)
scheduler_system.py:35:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
scheduler_system.py:35:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
scheduler_system.py:39:0: C0116: Missing function or method docstring (missing-function-docstring)
scheduler_system.py:39:0: C0103: Function name "logConflicts" doesn't conform to snake_case naming style
    (invalid-name)
scheduler_system.py:40:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
scheduler_system.py:40:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
scheduler_system.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)
scheduler_system.py:46:4: W0612: Unused variable 'a' (unused-variable)
scheduler_system.py:47:4: W0612: Unused variable 'a_more' (unused-variable)
scheduler_system.py:49:4: W0612: Unused variable 'c' (unused-variable)

-------------------------------------------------------------------
Your code has been rated at 5.60/10 (previous run: 5.60/10, +0.00)
```

After

After Pylint report

```
PS> pylint scheduler_system.py


-------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```

Bandit screenshots

Before

Before Bandit report

```
PS> bandit scheduler_system.py

[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.12.14
Run started:2026-10-14 06:24:39.250362+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\.\scheduler_system.py:27:4
26              del reservations[reservation_id]
27          except:
28              pass
29

--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injection detected,
    security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: .\.\scheduler_system.py:43:4
42              f.write(str(datetime.now()) + " " + c + "\n")
43          os.system("cat " + file)
44
```

After

After Bandit report

```
PS> bandit scheduler_system.py

[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.12.14
Run started:2026-10-14 06:26:52.817628+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 96
        Total lines skipped (#nosec): 0
        Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

Flake8 screenshots

Before

Before Flake8 report

```
PS> flake8 scheduler_system.py

.\scheduler_system.py:11:1: E302 expected 2 blank lines, found 1
.\scheduler_system.py:14:80: E501 line too long (87 > 79 characters)
.\scheduler_system.py:18:80: E501 line too long (102 > 79 characters)
.\scheduler_system.py:22:1: E302 expected 2 blank lines, found 1
.\scheduler_system.py:27:5: E722 do not use bare 'except'
.\scheduler_system.py:30:1: E302 expected 2 blank lines, found 1
.\scheduler_system.py:34:1: E302 expected 2 blank lines, found 1
.\scheduler_system.py:39:1: E302 expected 2 blank lines, found 1
.\scheduler_system.py:45:1: E302 expected 2 blank lines, found 1
.\scheduler_system.py:46:5: F841 local variable 'a' is assigned to but never used
.\scheduler_system.py:47:5: F841 local variable 'a_more' is assigned to but never used
.\scheduler_system.py:49:5: F841 local variable 'c' is assigned to but never used
.\scheduler_system.py:59:1: E305 expected 2 blank lines after class or function definition, found 1
```

After

After Flake8 report

```
PS> flake8 scheduler_system.py

No violations found.
```

Program run: before changes

```
Slot length for b: -2
'cat' is not recognized as an internal or external command,
operable program or batch file.
Slot for Board-2, like 10-12 (Enter for 10-12): Equipment left: {'projector': 1, 'microphone': 4, 'webcam': 0}
Reservations: {1: {'room': 'Board-1', 'slot': '09-11', 'equipment': []}, 2: {'room': 'Board-1', 'slot':
    '09-11', 'equipment': []}, 3: {'room': 'Huddle-3', 'slot': '14-12', 'equipment': []}, 4: {'room':
    'Huddle-3', 'slot': 1530, 'equipment': []}, 5: {'room': 'Board-2', 'slot': '10-12', 'equipment':
    ['projector', 'webcam']}}
```

The starter accepted the reversed slot 14-12 and reported -2 hours, stored an integer as a slot, and attempted to run the Unix `cat` command on Windows.

Program run: after changes

```
Invalid booking rejected: A time slot must end after it starts in one day.
Slot for Board-2, like 10-12 (Enter for 10-12): 2026-10-14 12:12:16.075200 Board-1 double-booked at 09-11
Equipment left: {'projector': 1, 'microphone': 4, 'webcam': 0}
Reservations: {1: {'room': 'Board-1', 'slot': '09-11', 'equipment': []}, 2: {'room': 'Board-1', 'slot':
    '09-11', 'equipment': []}, 3: {'room': 'Board-2', 'slot': '10-12', 'equipment': ['projector', 'webcam']}}
```

Behavior checks

```
PASS: unavailable equipment is rejected
PASS: cancellation restores stock and missing IDs are handled
PASS: reversed time slot is rejected
PASS: JSON export is valid and reservation IDs advance correctly
```

The checks covered valid and reversed time slots, equipment allocation and rejection, cancellation and stock restoration, reservation IDs, and JSON export.

Findings and fixes

The table prioritizes the high-severity Bandit issue, then the remaining scanner and manual-review findings. Line references refer to the starter file.

Issue	Tool / severity	Line	Finding and fix
Shell execution from filename	Bandit B605 / High	43	os.system("cat " + file) could pass filename text to a shell and failed on Windows. Replaced it with pathlib file reading using UTF-8.
Silent broad exception	Bandit B110 / Low; Pylint	27	Bare except/pass hid cancellation errors. Added an explicit missing-ID check and a clear False result; restore stock only for an existing booking.
Mutable default list	Pylint W0102	11	The default list could be reused across calls. Changed the default to None and create a fresh list for each booking.
Unmanaged file handles	Pylint R1732/W1514	35, 40	Files were opened without context managers or explicit encoding. Added with blocks and UTF-8.
Invalid slots accepted	Manual review	30-33, 48-50	Reversed times yielded negative durations and integer slots were stored. Validate format, bounds, and ordering before booking.
Global mutable state	Pylint W0603	5-9, 12	Global dictionaries and ID tracking made schedules hard to isolate. Moved state into a Scheduler instance.
Unchecked equipment quantity	Manual review	16-18	Inventory could go negative. Count and validate requests before changing stock.
Unused locals and style violations	Pylint / Flake8	11-59	Removed unused assignments, improved naming and spacing, and fixed the remaining line-length and exception-style violations.

Reflection

The handout says to answer four reflection questions but does not provide their wording. The responses below address four common prompts aligned with the lab objectives.

1. What issues did the analysis tools reveal?

Bandit found shell execution built from a filename and a broad exception that silently discarded errors. Pylint highlighted a shared mutable default, global state, missing documentation, unmanaged file handles, and unused variables. Flake8 found spacing, line-length, and exception-style violations. Manual execution exposed reversed and malformed slots being accepted.

2. How did fixing the issues improve the program?

The scheduler validates time slots and equipment before changing state, isolates each schedule in a Scheduler object, restores stock during cancellation, and reports missing IDs clearly. File operations use context managers and UTF-8, and logs are read without launching a shell.

3. Which tool was most useful, and why?

Pylint provided the broadest view of maintainability and a score to track progress. Bandit identified the shell-injection risk, while Flake8 gave a quick check of Python formatting.

4. What was challenging, and what did you learn?

Some behavior defects, including reversed time slots and equipment overbooking, required manual review in addition to scanner output. I learned to treat static analysis as a review aid: inspect findings, fix the underlying cause, then rerun the tools and program.